

Higher Criticism for Adaptive Retrieval in Retrieval-Augmented Generation

Lihu Zur and Noam Delbari

Final Project Report

lihu.zur@example.com noam.delbari@example.com

Abstract

Retrieval-Augmented Generation (RAG) pipelines typically retrieve a fixed number of passages per query, wasting context budget on easy questions and starving hard ones. We study *Higher Criticism* (HC), a statistical test for rare and weak signals, as a query-adaptive cutoff mechanism layered on top of any dense retriever. We integrate HC into three widely used RAG pipelines (Adaptive-K, Adaptive-RAG, FlashRAG) and into our own end-to-end evaluation pipeline, calibrate a per-corpus null distribution offline, and compare HC against fixed Top-K, Adaptive-K, and the multi-step Adaptive-RAG router across five evaluation datasets, with an additional focused failure-analysis study on BEIR FiQA. We also introduce *CrossEntityQA*, a sparse multi-entity retrieval benchmark in which only $\sim 2\%$ of the corpus is relevant per query. HC matches or beats fixed Top-K on Exact Match while using up to 95% fewer input tokens than Adaptive-K, dominates on long-context database-style reasoning (HoloBench), but underperforms on dependency-heavy multi-hop questions (MuSiQue, HotpotQA vs. multi-step IRCot). On controlled and compound-query benchmarks (Category-Key, Amazon Multi-Department / Electronics / Books) HC also beats every fixed top- k on F1 and dominates five alternative adaptive cut-off rules (Berk-Jones, Bonferroni, BH, gap-based Adaptive-K, clustering-based CAR). We characterise when HC helps, when it fails, and outline a hybrid rerank-plus-HC direction.

1 Background and Motivation

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), augments a Large Language Model (LLM) with passages pulled from an external corpus, typically via dense similarity search over an approximate-nearest-neighbour index. The dominant interface is *Top-K*: fix an integer k and pass the k most similar passages to the generator. This choice is convenient but blunt — a single k cannot simultaneously serve a fact-lookup query that needs one passage, a multi-hop question that needs three, and a database-style aggregation that needs dozens. Too small a k discards evidence; too large a k inflates context length,

latency, and dollar cost, and degrades generation quality through distractor noise.

1.1 Existing Adaptive Approaches

Two families of remedies have been proposed. *Adaptive-K* (Labs 2025) chooses k per query by locating the largest gap in the sorted similarity scores. *Adaptive-RAG* (Jeong et al. 2024) trains a small classifier that routes each query to one of {no retrieval, single-step retrieval, iterative IRCot}. The first is cheap but brittle when score distributions are flat; the second is accurate on multi-hop questions but pays for it with many retriever and generator calls.

1.2 Higher Criticism as a Cutoff Rule

Higher Criticism (HC) (Donoho and Jin 2004) is a goodness-of-fit statistic designed to detect a small number of *rare and weak* signals buried in a sea of nulls — exactly the statistical regime of a sparse-relevance retrieval ranking. Given a per-query score distribution and an offline null distribution of irrelevant-passage scores, HC ranks candidates by their z -score-based p -values and selects the cutoff that maximises the standardised deviation between observed and expected uniform p -values. The output is an adaptive k^* tied directly to the statistical separation of relevant from irrelevant passages.

1.3 Research Question

Can an HC-driven cutoff match or beat fixed- k and existing adaptive baselines on answer quality while substantially reducing context cost, and on which kinds of questions does it fail? This report answers that question across five datasets spanning single-hop QA, multi-hop QA, long-context aggregation, and scientific fact-checking, and introduces a purpose-built benchmark (*CrossEntityQA*) to stress-test the sparse-relevance regime that HC theory targets.

2 Methodology

2.1 HC Retrieval Algorithm

For each query q the retriever returns the top N candidates with cosine similarities $s_1 \geq s_2 \geq \dots \geq s_N$. Offline, we build a *null distribution* of similarities by sampling random (query, non-relevant passage) pairs from the corpus. At inference each s_i is converted to a z -score under the null, then

[Figure placeholder: HC-RAG architecture — offline indexing + HC calibration; online retrieve, cutoff, generate, evaluate.]

Figure 1: HC-RAG high-level architecture.

to a p -value p_i . Sorting $p_{(1)} \leq \dots \leq p_{(N)}$, the HC statistic is

$$HC_i = \sqrt{N} \frac{i/N - p_{(i)}}{\sqrt{p_{(i)}(1 - p_{(i)})}},$$

and the adaptive cutoff is $k^* = \arg \max_{i \leq \gamma N} HC_i$, with γ a small upper-fraction hyperparameter (typically 0.2). The pipeline returns the top- k^* passages.

2.2 System Architecture

Figure 1 shows the offline / online split. Offline: chunking, embedding, FAISS indexing, and null calibration. Online: query embedding, ANN retrieval, HC cutoff, prompt construction, generation, and joint IR + generation evaluation.

2.3 Pipelines Integrated

We integrated HC as a drop-in cutoff stage into:

- **Adaptive-K** (Labs 2025) — replaced the largest-gap heuristic with HC at the same insertion point; this method was designed for long-context QA in RAG.
- **Adaptive-RAG** (Jeong et al. 2024) — the classifier-routed framework that dispatches each query to one of {no-retrieval (NOR), single-step retrieval (OneR), iterative multi-hop retrieval (IRCoT)}; HC was inserted to size the retrieval at every IRCoT iteration.
- **FlashRAG** (Jin et al. 2025) — registered HC as a new retriever post-processor compatible with their sequential, conditional, branching, and iterative pipelines.
- **Our pipeline** — end-to-end RAG with configurable HuggingFace embedders, FAISS index, multiple chunking strategies, and OpenRouter for LLM generation; this is where all controlled HC vs. baseline experiments are run.

2.4 Additional Cutoff Baselines

For the algorithm comparison study (Section 5.3) we also evaluate four further selection rules that consume the same p -values as HC: **Berk-Jones** (Berk and Jones 1979), which replaces HC’s numerator with a Bernoulli KL divergence; **Bonferroni** (Dunn 1961) FWER control; **Benjamini-Hochberg** (Benjamini and Hochberg 1995) step-up FDR control; and **CAR** (Xu et al. 2025), a clustering-based cutoff operating on the similarity vector directly.

2.5 CrossEntityQA: A New Benchmark

Most public RAG benchmarks favour single-passage answers, which masks the regime HC targets. We constructed **CrossEntityQA**, a synthetic QA retrieval benchmark built from Wikipedia. The construction pipeline (Figure 2) clusters Wikidata entities into constraint-aligned groups of target

[Figure placeholder: CrossEntityQA construction pipeline (Wikidata → entity clusters → K -aligned subclusters → LLM query generation + filtering → corpus / queries / qrels).]

Figure 2: CrossEntityQA construction pipeline.

size $K \in \{2, \dots, 19\}$, attaches a Wikipedia passage per entity, and uses an LLM to write a query that is answerable only when all K passages are retrieved. The resulting corpus is *sparse*: $\sim 2\%$ of passages are relevant per query, with an average of 8.8 ground-truth passages spanning unrelated topic groups.

2.6 Datasets

We evaluate on five datasets spanning the regimes that stress an adaptive cutoff: **CrossEntityQA** (our sparse multi-entity benchmark), **HotpotQA** (Yang et al. 2018) (multi-hop QA), **MuSiQue** (Trivedi et al. 2022) (compositional multi-hop QA), **HoloBench** (Maekawa and others 2024) (long-context database-style aggregation), and **BEIR SciFact** (Thakur et al. 2021; Wadden et al. 2020) (scientific fact verification). For failure analysis (Section 4.5) we additionally use **BEIR FiQA** (Thakur et al. 2021).

The additional empirical studies in Section 5 exercise HC on three benchmarks designed in-house: **Category-Key** (44 “largest cities in X ” country queries over 726 Wikipedia passages, with variable ground-truth size $K \in [4, 24]$); and three compound-query corpora derived from McAuley-Lab/Amazon-Reviews-2023 — **Amazon Multi-Department** (80K docs, 135 subcategory+price queries across three departments), **Amazon Electronics** (100K, 44 queries), and **Amazon Book Categories** (100K, 131 queries).

2.7 Experimental Setup

Embeddings use BAAI bge-base-en-v1.5 (Xiao et al. 2023) (and bge-large-en-v1.5 for ablations). Documents are chunked with RecursiveChunker at 384 tokens / 50 overlap and indexed in FAISS (Johnson, Douze, and Jégou 2019). Generation uses gpt-4o-mini via OpenRouter with a fixed task-specific prompt template. HC uses an empirical null estimated from 10K random pairs; we tested parametric, rank-based, and position-aware nulls (Section 4.5) and retain the empirical null as default.

2.8 Metrics

Retrieval: Recall@ k , Precision@ k , average context size (avg k^*). Generation: Exact Match (EM), token-level F1, Precision, Recall. Efficiency: input tokens, average retrieved documents, end-to-end retrieval latency (seconds).

3 Contributions

1. **HC pipeline integrations** — Drop-in HC cutoff in Adaptive-K, Adaptive-RAG, FlashRAG, and a custom end-to-end pipeline.

Dataset / Metric	Top-K	HC	Δ
CrossEntityQA EM	0.500	0.751	+50%
CrossEntityQA Precision	0.670	0.711	+6.1%
CrossEntityQA Recall	0.610	0.572	-6.5%
MuSiQue EM	0.561	0.453	-19.2%
MuSiQue Precision	0.542	0.611	+12.7%
MuSiQue Recall	0.550	0.405	-26.4%
MuSiQue F1	0.546	0.487	-10.8%

Table 1: HC vs. Top-K ($k = 15$). HC wins on sparse single-hop QA, loses on dependency-heavy multi-hop QA.

2. **Null calibration study** — Empirical, parametric, rank-based, and position-aware null distributions, with evidence that the empirical null is sufficient and that per-query KS failures do not predict HC failures.
3. **CrossEntityQA benchmark** — A new sparse multi-entity retrieval dataset with variable ground-truth size ($K = 2-19$), designed to stress-test adaptive- k algorithms.
4. **Head-to-head evaluation** — Five datasets, three baselines (Top-K, Adaptive-K, Adaptive-RAG/IRCoT), reporting both quality and cost.
5. **Negative results and root-cause analysis** — We characterise the two failure modes (scattered relevance, multi-hop dependency) and a rerank-plus-threshold workaround that recovers +7.5% recall / +81% precision on FiQA.

4 Experiments and Results

4.1 HC vs. Top-K Baseline

Table 1 summarises HC vs. fixed Top-K ($k = 15$). On the sparse single-hop CrossEntityQA, HC lifts EM by +50% relative ($0.500 \rightarrow 0.751$) and precision by +6.1%, at a small recall cost. On the dependency-heavy MuSiQue, the picture inverts: HC retrieves a tighter set (precision 0.611 vs. 0.542) but loses the cross-passage evidence chain needed to answer (-19% EM).

4.2 HC vs. Adaptive-K

On HoloBench (long-context database-style aggregation) HC dominates Adaptive-K on every quality metric while using fewer documents and tokens: precision 0.890 vs. 0.531 (+67.6% relative), F1 0.706 vs. 0.497 (+42.1%), EM 0.374 vs. 0.326 (+14.8%), with a small recall trade-off (0.606 vs. 0.685). On HotpotQA the precision gap is even larger: HC doubles Adaptive-K’s precision (0.562 vs. 0.281; +100% relative) and lifts F1 by +60% (0.541 vs. 0.338) and EM by +23% (0.416 vs. 0.338). On BEIR SciFact HC adds +10% precision and +4% recall over Adaptive-K.

4.3 HC vs. Adaptive-RAG (IRCoT)

Against the multi-step Adaptive-RAG router on HotpotQA, HC underperforms by a uniform single-digit margin: EM 0.416 vs. 0.446 (-6.7%), precision 0.562 vs. 0.624 (-9.9%), F1 0.541 vs. 0.604 (-10.4%). IRCoT’s iterative query reformulation recovers cross-document evidence that a single-shot HC cutoff cannot, at the cost discussed below.

Dataset	Metric	HC	Adaptive-K
HotpotQA	Input tokens	1,320.2	25,249.9
HotpotQA	Avg # docs	10.9	227.6
HotpotQA	Retrieval lat. (s)	15.1	154.9
HoloBench	Input tokens	5,849	20,234
HoloBench	Avg # docs	179.1	638.4
HoloBench	Retrieval lat. (s)	1.46	11.49

Table 2: Efficiency: HC vs. Adaptive-K.

4.4 Efficiency

Table 2 shows the cost side. On HotpotQA HC uses 94.77% fewer input tokens (1,320.2 vs. 25,249.9), retrieves $\sim 21\times$ fewer documents (10.9 vs. 227.6), and is 90.25% faster (15.1 s vs. 154.9 s end-to-end). On HoloBench the gains carry over: 71% fewer tokens (5,849 vs. 20,234), $\sim 3.6\times$ fewer documents (179.1 vs. 638.4), and $\sim 8\times$ faster (1.46 s vs. 11.49 s). This is the central practical pitch for HC: when it matches or beats baselines on quality, it does so at a fraction of the dollar and latency budget.

4.5 Failure Analysis

To understand when HC fails we ran a focused position study on BEIR FiQA. HC implicitly assumes that relevant passages *cluster* at the top of the ranked list; the test then detects the boundary where the score distribution stops being unusual. On FiQA the relevant passages are instead *scattered*: only 6/100 queries have ≥ 2 consecutive relevant passages at the top, and the median position of a relevant passage is 5. Even after cross-encoder reranking (which lifts the bi-encoder Cohen’s d from 0.01 to 1.13) clustering improves only marginally (to 8/100).

A simple rerank-plus-fixed-threshold rule recovers most of what HC loses: $\text{score} \geq 0$ after cross-encoder rerank delivers +7.5% recall and +81% precision over the Top-5 baseline, suggesting a productive hybrid (Section 7).

5 Additional Empirical Studies

This section reports three studies that broaden the empirical picture beyond the long-form QA benchmarks above: a controlled- K benchmark designed to isolate HC’s adaptivity, a large-corpus compound-query retrieval study with end-to-end generation, and a head-to-head comparison of HC against five alternative adaptive cut-off rules.

5.1 Controlled- K Benchmark: Category-Key

On Category-Key (44 “largest cities in X ” queries over 726 Wikipedia passages; BGE bi-encoder, FAISS), HC achieves F1 0.893 (Recall 0.852, Precision 0.938) at mean $k^* = 9.7$, beating every fixed top- k in the sweep $k \in \{5, 8, 10, 12, 15, 20\}$ (best fixed F1 = 0.783 at $k=10$). The Pearson correlation between HC’s adaptive k^* and the per-query ground-truth K is 0.835 — direct evidence that the HC argmax tracks the true relevance boundary rather than collapsing to a constant.

Method	F1	Mean k
HC + rescale	0.209	34.0
Berk-Jones	0.202	34.8
Top-50	0.187	50.0
Top-20	0.185	20.0
Adaptive-K (gap)	0.138	9.0
CAR (cluster)	0.110	416.5
BH	0.074	1.6
Bonferroni	0.022	0.3

Table 3: Adaptive cut-off comparison on Amazon Multi-Department (135 compound queries, pool $N=1000$). HC posts the best F1; flat-threshold rules under-retrieve.

5.2 Compound-Query Retrieval: Amazon (80K–100K)

The Amazon benchmarks stress HC on heterogeneous product embeddings (OpenAI `text-embedding-3-small`, FAISS) and compound subcategory+price queries that standard embeddings encode only weakly. On the 80K **Multi-Department** corpus (135 queries spanning Clothing, Electronics, Home & Kitchen), HC reaches retrieval F1 0.204 at mean $k^* = 34.1$, ahead of Top-20 (0.185; +9.7%), Top-29 set to the average number of relevant documents per query (0.192; +6.3%), and Top-50 (0.187; +8.9%), while retrieving 32% fewer documents than Top-50. End-to-end with a `gpt-5-mini` generator HC posts Token-F1 0.299 vs. 0.261 for Top-20 (+14.8%), with an LLM-judge Answer Faithfulness near 5.0 (the maximum) across all methods, indicating the generator already filters distractors effectively. The same pattern holds on **Electronics** (HC F1 0.192 vs. Top-20’s 0.185; +3.5%) and **Book Categories** (HC F1 0.133 vs. Top-50’s 0.128; +4.0%, with 37% fewer documents).

5.3 Algorithm Comparison: HC vs. Other Adaptive Cut-offs

On the same 135 Amazon Multi-Department queries we compared HC head-to-head against five further adaptive cut-off rules (Table 3), all sharing an identical FAISS pool of $N=1000$ candidates. HC wins on F1 (0.209), with Berk-Jones a close second (0.202); the gap-based Adaptive-K (Labs 2025) sits well behind (0.138); the clustering-based CAR (Xu et al. 2025) collapses at its default pool (0.110) and matches HC only after pool retuning. The classical multiple-testing rules under-retrieve drastically: Bonferroni (0.022) and Benjamini-Hochberg (0.074) reject almost no candidates at a flat significance threshold, confirming that an *adaptive* test statistic (HC, BJ) is essential when per-query signal strength varies.

6 Discussion

When HC wins. Sparse-relevance corpora with well-separated embeddings and clustered relevant passages (CrossEntityQA, HoloBench, BEIR SciFact, Category-Key, Amazon Multi-Department). Here HC’s adaptivity translates directly into precision gains and large token savings.

When HC loses. Two regimes. (i) *Dependency-heavy multi-hop* questions (MuSiQue, HotpotQA vs. IRCOT) where the answer requires evidence that an iterative reformulation can find but a single cutoff cannot. (ii) *Scattered relevance* corpora (FiQA) where the clustering assumption is violated.

Response to presentation feedback. Feedback from the final presentation centred on four points, which we have addressed in this report: (i) null-calibration choices are now made explicit in the Methodology, with the empirical null retained after comparison against parametric, rank-based, and position-aware alternatives; (ii) the failure-mode taxonomy (scattered relevance vs. multi-hop dependency) is stated directly in the abstract and revisited above; (iii) the rerank-plus-threshold hybrid is reported as an explicit comparison point in Section 4.5 rather than only as future work; and (iv) efficiency numbers (tokens, document counts, latency) are reported alongside every quality result so that gains and trade-offs are directly comparable.

7 Conclusions and Future Work

We showed that Higher Criticism is a competitive query-adaptive cutoff for RAG: matching or beating fixed Top-K and Adaptive-K on four of five long-form QA datasets while reducing input-token cost by up to $\sim 95\%$; on the controlled Category-Key benchmark HC also beats every fixed top- k in the sweep, and on the Amazon compound-query benchmark it leads five alternative adaptive cut-off rules including Berk-Jones, BH, Bonferroni, gap-based Adaptive-K, and clustering-based CAR. We documented the two regimes where HC fails. Three directions stand out for future work:

- **Rerank + HC hybrid.** Apply HC after a cross-encoder rerank to address the scattered-relevance failure mode; the FiQA threshold experiment is a positive existence proof.
- **Iterative HC.** Embed HC inside Adaptive-RAG’s IRCOT loop to combine multi-hop coverage with adaptive-cost control.
- **Position-aware nulls.** Replace the per-query empirical null with a position-conditioned null to reduce variance without destroying HC’s signal-detection power.

References

- Benjamini, Y., and Hochberg, Y. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)* 57(1):289–300.
- Berk, R. H., and Jones, D. H. 1979. Goodness-of-fit test statistics that dominate the Kolmogorov statistics. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 47(1):47–59.
- Donoho, D., and Jin, J. 2004. Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics* 32(3):962–994.
- Dunn, O. J. 1961. Multiple comparisons among means. *Journal of the American Statistical Association* 56(293):52–64.

- Jeong, S.; Baek, J.; Cho, S.; Hwang, S. J.; and Park, J. C. 2024. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of NAACL*.
- Jin, J.; Zhu, Y.; Yang, X.; Zhang, C.; and Dou, Z. 2025. Flashrag: A modular toolkit for efficient retrieval-augmented generation research. In *Proceedings of the ACM Web Conference (WWW)*.
- Johnson, J.; Douze, M.; and Jégou, H. 2019. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*.
- Labs, M. 2025. Adaptive-k retrieval for long-context question answering. In *Proceedings of EMNLP*.
- Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.-t.; Rocktäschel, T.; Riedel, S.; and Kiela, D. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Proceedings of NeurIPS*.
- Maekawa, S., et al. 2024. Holobench: A benchmark for holistic long-context reasoning over text. In *Proceedings of NeurIPS Datasets and Benchmarks*.
- Meta AI. 2024. CRAG: Comprehensive RAG benchmark. <https://github.com/facebookresearch/CRAG>.
- Thakur, N.; Reimers, N.; Rücklé, A.; Srivastava, A.; and Gurevych, I. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of NeurIPS Datasets and Benchmarks*.
- Trivedi, H.; Balasubramanian, N.; Khot, T.; and Sabharwal, A. 2022. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics* 10:539–554.
- Wadden, D.; Lin, S.; Lo, K.; Wang, L. L.; van Zuylen, M.; Cohan, A.; and Hajishirzi, H. 2020. Fact or fiction: Verifying scientific claims. *Proceedings of EMNLP*.
- Xiao, S.; Liu, Z.; Zhang, P.; and Muennighoff, N. 2023. C-Pack: Packaged resources to advance general chinese embedding. <https://huggingface.co/BAAI/bge-base-en-v1.5>.
- Xu, Y.; Gupta, A.; Aggarwal, C.; Mahadevan, S.; and Krishnamachari, B. 2025. Cluster-adaptive retrieval (CAR) for adaptive- k selection. arXiv:2511.14769.
- Yang, Z.; Qi, P.; Zhang, S.; Bengio, Y.; Cohen, W. W.; Salakhutdinov, R.; and Manning, C. D. 2018. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of EMNLP*.

Appendix A: Project Roadmap

Week	Milestone / Outcome
1–2	CRAG (Meta AI 2024) dataset acquisition; baseline RAG with <code>all-mpnet-base-v2</code> ; full-document retrieval.
3	<code>RecursiveChunker</code> (384 / 50 overlap); k -sweep on 500-doc subset — $k=10$ optimal (R@k 0.817, P@k 0.503).
4	BGE <code>bge-base-en-v1.5</code> on 1K docs (R@k 0.826); selected as default embedder.
5	First HC integration into our pipeline; per-query null calibration on CRAG.
6	End-to-end evaluation on BEIR FiQA, 648 queries with <code>gpt-4o-mini</code> : HC ties baseline at 52% more docs — first negative result.
7	Null-distribution study: empirical vs. parametric vs. rank-based vs. position-aware; empirical retained.
8	Failure root-cause: FiQA relevant passages are scattered (0.49 mean consecutive at top), not clustered.
9	Cross-encoder rerank + threshold workaround: +7.5% recall, +81% precision over Top-5.
10	CrossEntityQA construction pipeline (Wikidata + Wikipedia + LLM-generated queries).
11	HC integration into Adaptive-K, Adaptive-RAG, FlashRAG.
12	Comparative evaluation on HotpotQA, HoloBench, MuSiQue, BEIR SciFact, CrossEntityQA; efficiency benchmarks.
13	Final presentation; collected class feedback.
14	Report draft; additional experiments addressing feedback (in progress).